

Omid Hazara

Software Engineer | Java | Python | Backend & AI Systems

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SUMMARY

Software engineer with 3+ years building backend services, system integrations, and data migration tooling in enterprise Product Lifecycle Management (PLM) environments. MSc in Computer Engineering from KTH, machine learning focus. Java and Python primary, with production C#/.NET and AWS experience.

EXPERIENCE

Software Engineer | Scania CV AB, Stockholm | 2023 - Present

- Built end to end, from design through deployment, a C#/.NET API for automated PDF generation in the 3DEXPERIENCE PLM platform within the first six months in role, documented with OpenAPI/Swagger. Handles several thousand documents per day for 800+ users at around 99% fidelity, under 5 needing manual follow-up.
- Design, investigate and build secure, scalable server-side solutions and integrations in ENOVIA, driving product data handoffs to ERP, BOM, CAD and manufacturing systems through event-triggered and lifecycle-driven workflows.
- Collaborate on a large-scale migration between legacy CATIA V5 / 3DEXPERIENCE environments and the current platform, contributing across an extract, transform and load flow on Apache Ant, XSLT and XML parsing with YAML-driven config that moves about 10,000 objects weekly.
- Develop custom 3DEXPERIENCE widgets on the UWA framework using Vue.js, delivering in-platform user interfaces for internal tools.
- Designing and building a multipurpose internal AI assistant (PoC) giving engineers, offshore support and back-office staff self-service access to process knowledge and live platform data: RAG over process documentation answers review and approval workflow queries, while MCP tool calling retrieves object state from the PLM platform. LangChain, AWS, Vue.js, self-hosted LLM.
- Mentor and onboard engineers joining the organisation (since September 2025).

Engineering Intern & Thesis Worker | Scania CV AB, Stockholm | 2019 - 2023

- Master's thesis: built a 3D model similarity search prototype using CNN-based embeddings, evaluated over a catalogue of 1,100 models, helping engineers find reusable geometry instead of modelling from scratch.
- R&D IT internship: co-built a parser for a CO2 audit tool backed by a knowledge graph, ingesting REST APIs and transforming XML into RDF via RDF4J for SPARQL querying on Stardog and Apache Jena Fuseki, with ontology modelling and Dockerised deployment.
- Automated recurring internal business processes using RPA, and built Power BI dashboards for staff-division management.
- Maintained and extended a large legacy Excel/VBA integration layer fetching and processing data from multiple R&D endpoints to produce Gantt-based project timelines and status reporting.

PROJECTS

Ardent: Multi-Agent AI Platform | Personal project | Ongoing

- Architecting a multi-agent orchestration platform on LangGraph with MCP tool integration, LiteLLM as model gateway, and vLLM for self-hosted inference.
- Hybrid knowledge base combining Qdrant vector storage (BGE-M3 embeddings), PostgreSQL, and Neo4j for graph relations, backing a RAG pipeline over domain documents.
- Instrumented agent runs with Langfuse for tracing and evaluation; architecture documented as ADRs.

EDUCATION

MSc, Computer Engineering | KTH Royal Institute of Technology, Stockholm | Graduated 2024

Algorithms & Data Structures, Machine Learning, Distributed Data Processing (Apache Spark), Software Engineering, Cybersecurity, DevOps.

Exchange Studies | University of Illinois Urbana-Champaign | Spring 2022

SKILLS

Programming Languages: Java, Python, C#/.NET, JavaScript/TypeScript, SQL, Bash

Backend & Frameworks: ASP.NET, FastAPI, REST API design, OpenAPI/Swagger, Vue.js, React

Data & Integration: PostgreSQL (pgvector), Qdrant, MongoDB, XML/XSLT, Ant, ETL

Knowledge Graphs: RDF, SPARQL, ontology modelling, Jena Fuseki, Neo4j

AI & ML: LangChain, LangGraph, MCP, RAG, vLLM, LiteLLM, Ollama, Langfuse, PyTorch, NumPy, pandas, scikit-learn

Cloud & DevOps: AWS, Terraform, Docker, Git, GitLab CI/CD, JFrog Artifactory, Keycloak / Entra ID

Enterprise Platforms: 3DEXPERIENCE / ENOVIA (MQL, JPO, UWA widgets), CATIA V5, Power BI, RPA

Spoken Languages: Persian (native), Swedish and English (fluent), German (intermediate)